

U.G.C. NET Exam

COMPUTER

(Solved with Expl)

1. Consider a set $A = \{1, 2, 3, \dots, 1000\}$. How many members of A shall be divisible by 3 or by 5 or both 3 and 5?

(a) 533 (b) 599
(c) 467 (d) 66

Ans : (c) Because, number of member divisible by 3 = $\{3, 6, 9, \dots, 999\} : 999/3 = 333$

Number of member divisible by 5 = $\{5, 10, 15, \dots, 1000\} : 1000/5 = 200$

Number of member divisible by 15 = $\{15, 30, \dots, 990\} : 999/15 = 66$

So, no. divisible by 3 or by 5 or both 3 and 5.

$$(A \cup B) = (A + B) - (A \cap B)$$

Thus the total number in set = $333 + 200 - 66 = 467$

So the answer is (c)

2. A certain tree has two vertices of degree 4, one vertex of degree 3 and one vertex of degree 2. If the other vertices have degree 1, how many vertices are there in the graph?

(a) 5 (b) $n-3$
(c) 20 (d) 11

Ans : (d) Number of edges in a tree with n vertices:

$$e = n - 1$$

Now, using theorem Handshaking theorem

$$\sum_{V \in V} \deg(V) = 2|e|$$

Therefore,

$$\Rightarrow 2 \times 4 + 1 \times 3 + 1 \times 2 + (n - (2 + 1 + 1)) \times 1$$

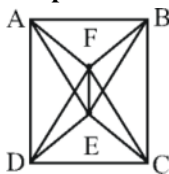
$$= 2(n - 1)$$

$$\Rightarrow 13 + (n - 4) = 2n - 2$$

$$\Rightarrow 13 + 2 - 4 = 2n - n$$

$$\Rightarrow n = 11$$

3. Consider the Graph shown below :



This graph is a

(a) Complete graph (b) Bipartite graph
(c) Hamiltonian graph (d) All of the above

Ans : (c) Because, According to Dirac's theorem, in a graph of n nodes, each node has degree greater than $n/2$, graph is Hamiltonian

- In given graph $n = 6$
- All nodes have degree = 4
- Hence graph is Hamiltonian
- So Hamiltonian path is ABCDEF

So the answer is C

n. December, 2014

R SCIENCE

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4. A computer program select an integer in the set $\{k : 1 \leq k \leq 10,00,000\}$ at random and prints out the result. This process is repeated 1 million times. What is the probability that the value $k = 1$ appears in the printout at least once?

- (a) 0.5 (b) 0.704
(c) 0.632121 (d) 0.68

Ans : (c) Because, probability of $k = 1$ is $1/1000000 = 10^{-6}$

probability of $k \neq 1$ is $1 - 10^{-6} = 0.999$

probability that $k = 1$ is never printed in all 106 print outs = $0.999 \times 0.999 \times \dots \dots \dots 0.999$ (10^6 time)

= $0.999 \cdot 10^6$

Probability that 1 is printed at least once = $1 -$ probability that 1 is never printed

$\Rightarrow 1 - 0.999^{10^6}$

$\Rightarrow 0.6321$

5. If we define the functions f, g and h that map R into R by :

$f(x) = x^4, g(x) = \sqrt{x^2 + 1}, h(x) = x^2 + 72$, then the value of the composite functions $ho(gof)$ and (hog) of are given as

- (a) $x^2 - 71$ and $x^8 - 71$
(b) $x^8 - 73$ and $x^8 - 73$
(c) $x^8 + 71$ and $x^8 + 71$
(d) $x^8 + 73$ and $x^8 + 73$

Ans : (d) $f = x^4$

$$gof = \sqrt{(x^8 + 1)}$$

$ho(gof) = x^8 + 1 + 72 = x^8 + 73$ // option d hold this!

$ho(gof) = x^8 + 73 = x^8 + 73$

So the answer is (d)

6. The BCD adder to add two decimal digits needs minimum of

- (a) 6 full adders and 2 half adders
(b) 5 full adders and 3 half adders
(c) 4 full adders and 3 half adders
(d) 5 full adders and 2 half adders

Ans : (d) Because, BCD adder to add two decimal digit needs 5 full adders & 2 half adders.

So answer is (d)

7. The Excess-3 decimal code is self-complementing code because

- (a) The binary sum of a code and its 9's complement is equal to 9
(b) It is a weighted code

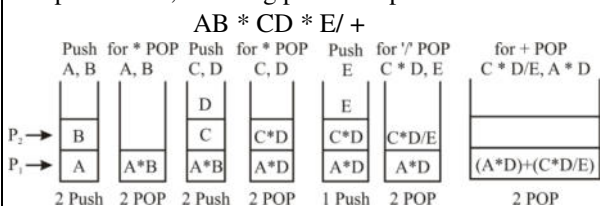
- (c) Complement can be generated by inverting each bit pattern
- (d) The binary sum of a code and its 10's complement is equal to 9

Ans : (a or c) The Excess-3 decimal code is a self-complementing code because the binary sum of a code and its 9's complement is equal to 9 and complement can be generated by inverting each bit pattern.

8. How many PUSH and POP operations will be needed to evaluate the following expression by reverse polish notation in a stack machine (A × B) + (C × D/E)?

- (a) 4 PUSH and 3 POP instructions
- (b) 5 PUSH and 4 POP instructions
- (c) 6 PUSH and 2 POP instructions
- (d) 5 PUSH and 3 POP instructions

Ans : (b) To evaluate postfix expression we need operand stack given expression is infix, so convert it into postfix. So, resulting postfix expression:



So, total 5 push and 9 POP. We can reduce POP since keep one operand in separate register, so need only 1 POP operation instead of 2 POP. So, can be answer but if 5 push and 8 POP given then 'b' is wrong.

9. The range of representable normalized numbers in the floating point binary fractional representation in a 32-bit word with 1-bit sign, 8-bit excess 128 biased exponent and 23-bit mantissa is :

- (a) 2^{-128} to $(1-2^{-23}) \times 2^{127}$
- (b) $(1-2^{-23}) \times 2^{-127}$ to 2^{128}
- (c) $(1-2^{-23}) \times 2^{-127}$ to 2^{23}
- (d) 2^{-129} to $(1-2^{-23}) \times 2^{127}$

Ans : (d) Because, the range of respectable normalized number in the floating point binary fractional representation in a 32-bit word

$$\text{bias} = 2^{n-1} = 2^{8-1} = 128$$

Number 1 in normalized format = 0.00000.....23 times
 $= 2^{0-128} = 2^{-128}$

Maximum number 11111.....22 time = 0.1111.....23 times
 $= (1 - 2^{-23}) \times (2^{555-228}) = (1 - 2^{-23}) \times (2^{127})$

So the answer is (d)

10. The size of the ROM required to build an 8-bit adder/subtractor with mode control, carry input, carry output and two's complement overflow output is given as :

- (a) $2^{16} \times 8$
- (b) $2^{18} \times 10$
- (c) $2^{16} \times 10$
- (d) $2^{18} \times 8$

14. When an array is passed as parameter to a function, which of the following statements is correct?

- (a) The function can change values in the original array
- (b) In C, parameters are passed by value, the function cannot change the original value in the array
- (c) It results in compilation error when the function tries to access the elements in the array
- (d) Results in a run time error when the function tries to access the elements in the array

Ans : (a) When an array is passed as parameter to a function then the function can change values in the original array.

15. Which of the following differentiates between overloaded functions and overridden functions?

- (a) Overloading is a dynamic or runtime binding and overridden is a static or compile time binding
- (b) Overloading is a static or compile time binding and overriding is dynamic or runtime binding
- (c) Redefining a function in a friend class is called overloading, while redefining a function in a derived class is called as overridden function
- (d) Redefining a function in a derived class is called function overloading, while redefining a function in a friend class is called function overriding

Ans : (b) Because, In method overloading, method call to its definition happens at compile time & while in method overriding, method call to its definition happens at run time.

So the answer is (b)

16. Division operation is ideally suited to handle queries of the type :

- (a) customers who have no account in any of the branches in Delhi
- (b) customers who have an account at all branches in Delhi
- (c) customers who have an account in atleast one branch in Delhi
- (d) customers who have only joint account in any one branch in Delhi

Ans : (b) Because division identifies the attribute values from relation that are found to be paired with all of the values from the other relation.

So the answer is (b)

17. Which of the following is true?

- I. Implementation of self-join is possible in SQL with table alias
- II. Outer-join operation is a basic operation in relational algebra

III. Natural join and outer join operations are equivalent.

- (a) I and II are correct
- (b) II and III are correct
- (c) Only III is correct
- (d) Only I is correct

Ans : (d) Because only I is correct. Self join is implemented in a single table by alias names rest 2 are incorrect, so the answer is (d)

18. What kind of mechanism is to be taken into account for converting a weak entity set into strong entity set in entity relationship diagram?

- (a) Generalization
- (b) Aggregation
- (c) Specialization
- (d) Adding suitable attributes

Ans : (d) Weak entity is an entity which does not have sufficient attributes to form a primary key. Weak entity is represented by Double rectangle In an ER Diagram. It must be in a 1:M relation. A member of a weak entity is called a subordinate entity. For converting a weak entity set into strong entity set in entity-relationship diagram by simply adding appropriate attributes.

19. The best normal form of relation scheme R(A, B, C, D) along with the set of functional dependencies $F = \{ AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B \}$ is :

- (a) Boyce-Codd Normal form
- (b) Third Normal form
- (c) Second Normal form
- (d) First Normal form

Ans : (b) Because, keys are AB, CD, BC, AD
Here A, B, C, D all are prime attributes
So, no partial or transitive dependency are there
So it in 3NF
Hence the answer is (b)

20. Identify the minimal key for relational scheme R(A, B, C, D, E) with functional dependencies $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$

- (a) A
- (b) AE
- (c) BE
- (d) CE

Ans : (b) Because, $\{A\}^+ = \{A, B, C, D\}$
Here A can not be a key because we can't get attribute E from here.
 $\{AE\}^+ = \{A, B, C, D, E\}$
Here AE is the minimal key we can't remove anything from it
Hence the correct choice is (b)

21. Convert the following infix expression into its equivalent post fix expression

$(A + B^D)/(E - F) + G$

- (a) $ABD^+ + EF -/G+$
- (b) $ABD+ ^EF -/G+$

- (c) $ABD + \wedge EF / -G +$ (d) $ABD^{\wedge} + EF / -G +$

Ans : (a) Given infix notation is

$$= (A + B \uparrow D) / (E - F) + G$$

= Convert it into postfix expression

$$= (((((A + (B \uparrow D)) / (E - F)) + G)$$

$$= (((((ABD \uparrow +) / (EF -)) + G)$$

$$= ABD \uparrow + EF - / G +$$

22. You have to sort a list L. consisting of a sorted list followed by a few 'random' elements. Which of the following sorting method would be most suitable for such a task?

- (a) Bubble sort (b) Selection sort
(c) Quick sort (d) Insertion sort

Ans : (d) For a sorted list with few random numbers:

(i) Bubble sort will take $O(n^2)$ time in best case.

(ii) Selection sort will take $O(n^2)$ time in best case.

(iii) Quick sort will take $O(n^2)$ time in this case because it is the worst case for quick sort.

(iv) Insertion sort will take $O(n)$

23. The directory can be viewed as that translates filenames into their directory entries.

- (a) Symbol table (b) Partition
(c) Swap space (d) Cache

Ans : (a) The directory can be viewed as symbol table that translates file-names into their directory entries. Symbol table is an important data structure created and maintained by compilers in order to store information about the occurrence of various entities such as variable names, function names, objects, classes, interfaces, etc.

24. Consider an array A[20, 10], assume 4 words per memory cell and the base address of array A is 100. What is the address of A [11, 5]? Assume row major storage.

- (a) 560 (b) 565
(c) 570 (d) 575

Ans : (a) Because, Each cell needs 4 words

- A [0][0], A[0] [1],..... A[0][9] needs $4 \times 10 = 40$ words
- Since array starts at 100, A [0] [9] will be at $100 + 40 = 140^{\text{th}}$ location
- Similarly from A[0][0] to A[10][9] there are $11 \times 10 = 110$ elements
- From A[11][0] to A[11][5] there are 5 elements
- Total 115 elements are present before A[11][15]
- Hence A[11][15] will be $100 + 4 \times 115 = 560$

So the answer is (a)

25. A full binary tree with n leaves contains

- (a) n nodes (b) $\log_2$ nodes
(c) $2n - 1$ nodes (d) 2^n nodes

Ans : (c) Because, full binary tree with n leaves contains $n-1$ internal nodes

$$\text{So total nodes} = \text{internal nodes} + \text{leaves} = n - 1 + n = 2n - 1$$

So the answer is (c)

26. The period of a signal is 10 ms. What is its frequency in Hertz?

- (a) 10 (b) 100
(c) 1000 (d) 10000

Ans : (b) Because, frequency = $1/\text{period} = 1/10 \text{ ms} = 1/10^{-2} \text{ s} = 100 \text{ Hz}$
So the answer is (b)

27. In a classful addressing, first four bits in Class A IP address is :

- (a) 1010 (b) 1100
(c) 1011 (d) 1110

Ans : (none of these) In class A the first bit (MSB) of first octet is 0.

Class A: 0xxx : x can be 0 or 1.

Class B: 10xx

Class C: 110x

Class D: 1110

Class E: 1111

28. Which of the following algorithms is not a broadcast routing algorithm?

- (a) Flooding
- (b) Multi destination routing
- (c) Reverse path forwarding
- (d) All of the above

Ans : (d) Because, flooding is suited for ordinary point to point communication, for broadcasting it might be good solution. The problem with flooding is it generates too many packets & consumes too much bandwidth. Multi destination routing where each packet contains either a list of destination and reverse path forwarding is an attempt to approximate the spanning tree broadcast even when the routers do not know anything at all about spanning tree.

So the correct choice is (d)

29. An analog signal has a bit rate of 6000 bps and a baud rate of 2000 baud. How many data elements are carried by each signal element?

- (a) 0.336 bits/ baud
(b) 3 bits/ baud
(c) 120,00,000 bits/ baud
(d) None of the above

Ans : (b) Because the analog signal 2000 bauds or element in one second.

- To send this 6000 bits are transmitted second
- Thus $6000/2000 = 3$ bits/ baud is transmitted

30. How many distinct stages are there in the DES algorithm, which is parameterized by a 56-bit key?

- (a) 16 (b) 17
(c) 18 (d) 19

Ans : (d) Because, DES (Data encryption Standard) algorithm which is parameterized by a 56 bit key has 19 distinct edges including 16 rounds.
So the answer is (d)

31. Shift-Reduce parsers perform the following :

- (a) Shift step that advances in the input stream by $K(K > 1)$ symbols and Reduce step that applies a completed grammar rule to some recent parse trees, joining them together as one tree with a new root symbol.
- (b) Shift step that advances in the input stream by one symbol and Reduce step that applies a completed grammar rule to some recent parse trees, joining them together as one tree with a new root symbol
- (c) Shift step that advances in the input stream by $K(K = 2)$ symbols and Reduce step that applies a completed grammar rule to form a single tree.
- (d) Shift step that does not advance in the input stream and Reduce step that applies a completed grammar rule to form a single tree

Ans : (b) Because, Shift step advances in the input stream by one symbol. That shifted symbol becomes a new single-node parse tree

- Reduce step applies a completed grammar rule to some of the recent parse tree, joining them together as one tree with a new root symbol.
- If the input has no syntax errors, the parser continues with these steps until all of the input has been consumed and all of the parse tree have been reduced to a single tree representing an entire legal input.

32. Which of the following is true?

- (a) Canonical LR parser is LR (1) parser with single look ahead terminal
- (b) All LR (K) parsers with $K > 1$ can be transformed into LR (1) parsers
- (c) Both (a) and (b)
- (d) None of the above

Ans : (c) Because, A canonical LR parser or LR (1) is an LR (k) parser for $k = 1$, i.e. with a single look ahead all LR (k) parser's with $k > 1$ can be transformed into a LR (1) parser.

33. In a two-pass assembler, symbol table is :

- (a) Generated in first pass
- (b) Generated in second pass
- (c) Not generated at all
- (d) Generated and used only in second pass

Ans : (a) In a two-pass assembler, symbol table is generated in first pass but it need two pass to scan whole source file. While one pass assembler scan whole file in a go.

34. Debugger is a program that :

- (a) allows to examine and modify the contents of registers
- (b) does not allow execution of a segment of program
- (c) allows to set breakpoints, execute a segment of program and display contents of register
- (d) All of the above

Ans : (c) Because, debugger is computer program used by programmers to test and debug a target program and it use instructions - set simulators rather than running a program directly on the processor to achieve a higher level of control over its execution. It also use to allow, set breakpoints, execute a segment of program and display content of register.

35. The following context-free Grammar (CFG) :

$S \rightarrow aB \mid bA$

$A \rightarrow a \mid as \mid bAA$

$B \rightarrow b \mid bs \mid aBB$

will generate

- (a) odd numbers of a's and odd numbers of b's
- (b) even numbers of a's and even numbers of b's
- (c) equal numbers of a's and b's
- (d) different numbers of a's and b's

Ans : (c) Because, if you consider this CFG you can definitely generate odd no. of a's and odd no. of b's and we can generate the string aabb (which is a contradiction of option b & a)

Given CFL also can generate ab(which is contradiction to the option d)

⇒ Means when you consider option (c) in this you can generate all possible string out of this grammar is which you are going to have equal no. of a's and equal no. of b's

⇒ So the correct choice is (c)

36. Consider the following justifications for commonly using the two-level CPU scheduling:

- I. It is used memory is too small to hold all the ready processes**
- II. Because its performance is same as that of the FIFO.**
- III. Because it facilitates putting some set of processes into memory and a choice is made from that.**
- IV. Because it does not allow to adjust the set of in-core processes**

Which of the following is true?

- (a) I, III and IV
- (b) I and II
- (c) III and IV
- (d) I and III

Ans : (d) Because, two level scheduling is a method to more efficiently perform process scheduling that involved swapped out process. Two level scheduling uses two different schedulers, one lower-level schedulers which can only select among those processes in memory to run. That scheduler could be a round Robin scheduler. The other scheduler is the higher level scheduler whose only concern is to swap in and swap out processes from memory.

37. A specific editor has 200 K of program text, 15 K of initial stack, 50 K of initialized data, and 70 K of bootstrap code. If five editors are started simultaneously, how much physical memory is needed if shared text is used?

- (a) 1135 K
- (b) 335 K
- (c) 1065 K
- (d) 320 K

Ans : (b) Because, initial stack, initialized data and bootstrap code can be shared among the editors.
Because text is shared so no need to multiply 200 with 5 and other need each for running. Then the total memory is = $200 + 15 + 150 + 70 = 335K$

38. Which of the following conditions does hold good for a solution to a critical section problem?

- (a) No assumptions may be made about speeds or the number of CPUs.
- (b) No two processes may be simultaneously inside their critical sections.
- (c) Processes running outside its critical section may block other processes
- (d) Processes do not wait forever to enter its critical section.

Ans : (c) Because, if a process is running outside the critical section we don't care whatever it is doing out there either it is blocking some processes or not it is not going to affect the processes which are running inside the critical section.

39. For the implementation of a paging scheme, suppose the average process size be x bytes, the page size be y bytes, and each entry requires z bytes. The optimum page size that minimizes the total overhead due to the page table and the internal fragmentation loss is given by :

- (a) $\frac{x}{2}$
- (b) $\frac{xz}{2}$
- (c) $\sqrt{2xz}$
- (d) $\frac{\sqrt{xz}}{2}$

Ans : (c) Because, average amount of internal fragmentation per segment is $y/2$

Average no. of pages per process segment is x/y each page requires 'z' bytes of page table.

So each process segment require page table size of xz/y bytes.

Total overload per segment, therefore, due to internal fragmentation to minimize the overhead, differentiate with respect to page size, P & equal to 0

$$d(xz/y + y/2)/dy = -xz/y^2 + 1/2 = 0$$

$$\Rightarrow y = \sqrt{2xz}$$

So the answer is (c)

40. In a demand paging memory system, page table is held in registers. The time taken to service a page fault is 8 m. sec. if an empty frame is available or if the replaced page is not modified, and it takes 20 m. sec. if the replaced page is modified. What is the average access time to service a page fault assuming that the page to be replaced is modified 70% of the time?

- (a) 11.6 m. sec.
- (b) 16.4 m. sec.
- (c) 28 m. sec.
- (d) 14 m. sec.

Ans : (b) Because service a page fault = 8 m.sec.
page is not modified = 20 m.sec.

page replace = 70% of time

then the average time of service a page fault

$$= 0.7 \times 20 + 0.3 \times 8 = 14.4 = 16.4 \text{ ms}$$

41. are applied throughout the software process.
- (a) Framework activities
 - (b) Umbrella activities
 - (c) Planning activities
 - (d) Construction activities

Ans : (b) The **umbrella activities** in a software development life cycle process include the following: software project management, formal technical reviews, software quality assurance and software configuration management.

Umbrella activities occur throughout the software process and focus primarily on project management, tracking and control.

42. Requirement Development, Organizational process focus, organizational training, risk management and integrated supplier management are process areas required to achieve maturity level
- (a) Performed
 - (b) Managed
 - (c) Defined
 - (d) Optimized

Ans : (c) Because, maturity level 3 – defined

ATM : Acquisition technical management
AVAL : Acquisition validation
IPM : Integrated project management
OPF : organizational process focus
OT : organizational training
RSKM : Risk management

43. The software of a program of a computing system is the structure or structures of the system, which comprise software components, the externally visible properties of those components, and the relationships among them.
- (a) Design
 - (b) Architecture
 - (c) Process
 - (d) Requirement

Ans : (b) Because, software architecture encompasses the set of significant decisions about the organization of a software system including the selection of the structural elements and their interfaces by which the system is composed; behavior as specified in collaboration among those elements, composition of these structural and behavioral elements into larger subsystems : software architecture also involves functionality, usability, performance etc.

44. Which one of the following set of attributes should not be encompassed by effective software metrics?
- (a) Simple and computable
 - (b) Consistent and objective
 - (c) Consistent in the use of units and dimensions
 - (d) Programming language dependent

Ans : (d) Because, effective software metrics should be simple, computable, consistent, objective and consistent in the use of units and dimensions. But it should not be programming language dependent as then it will not be consistent in measuring the software.

Hence the answer is option (d)

45. Which one of the following is used to compute cyclomatic complexity?

- (a) The number of regions–1
- (b) $E - N + 1$, where E is the number of flow graph edges and N is the number of flow graph nodes
- (c) $P - 1$, where P is the number of predicate nodes in the flow graph G .
- (d) $P + 1$, where P is the number of predicate nodes in the flow graph G .

Ans : (d) Because, cyclomatic complexity is a metric to measure logical complexity of a program.

- Cyclomatic complexity defines the number of independent paths in a program
- It gives an upper bound for the number of tests that must be conducted to ensure that all statements have been executed at least once.

⇒ Cyclomatic complexity is computed in one of three ways.

- (1) Number of regions of the flow graph
- (2) Cyclomatic complexity, $V(G)$, for a flow graph G , is defined as $V(G) = E - N + 2K$ where E is the number of flow graph edges, N is the number of flow graph nodes and K is the number of connected components
- (3) $V(G) = P + 1$ where P is the number of predicate nodes contain in the flow graph G . So the answer is (d)

46. Consider the following statements S1 and S2 :

S1 : A hard handover is one in which the channel in the source cell is retained and used for a while in parallel with the channel in the target cell.

S2 : A soft handover is one in which the channel in the source cell is released and only then the channel in the target cell is engaged.

- (a) S1 is true and S2 is not true.
- (b) S1 is not true and S2 is true.
- (c) Both S1 and S2 are true.
- (d) Both S1 and S2 are not true.

Ans : (d) Because, Hard handover is one in which the channel in the source cell is released and only the channel in the target cell is engaged

⇒ Soft handover is one in which the channel in the source cell is retained and used for a while parallel with the channel in the target cell

⇒ So the both are not true, hence the correct choice is (d)

47. Fact-less fact table in a data warehouse contains

- (a) only measures
- (b) only dimensions
- (c) keys and measures
- (d) only surrogate keys

Ans : (b) A fact-less fact table is a fact table that does not have any measures. It is essentially an intersection of dimensions (it contains nothing but dimensional key) there are two types of fact-less tables : one is for capturing an event and one is for describing conditions.

